Amirmohammad Mahdavikia

EDUCATION

Tehran University of Medical Sciences (TUMS) 2020 - Expected 2027

Doctor of Medicine (MD)

EXPERIENCE

Machine learning developer 2023 - Present

- Freelancer
- NLP | CV | VLMs

Biology teacher Sep 2020 - Dec 2022

- Biochemistry | Cellular & Molecular biology | Genetics | Biostatistics

PROJECTS

Stanford RNA 3D Folding | Kaggle

- Independently investigated state-of-the-art methods for RNA tertiary structure prediction by benchmarking and running inference with multiple CASP16-derived frameworks, including Protenix (AF3), DRFold2, NuFold, RhoFold+, DeepFoldRNA, and trRosettaRNA. Conducted an extensive literature and model survey spanning ab initio, template-based, and deep learning approaches, and designed a multi-model ensemble integrating inference and refinement pipelines for improved generalization. Explored fine-tuning of AlphaFold3-based models on newly released PDB structures and implemented sequence-chunking strategies for handling long RNAs under limited GPU resources. The project deepened my understanding of biomolecular folding, CASP methodology, and cross-domain applications of AI in structural biology.

SPIDER | RSNA2024 competition

- Designed a pipeline using nnUNet for automatic segmentation of lumbar spinal MRI images, handling 35GB of DICOM and NIfTI formats, with deep knowledge of metadata extraction. Created slice-based pipelines, merging different modality images for better spatial information extraction.

DarooGAP | HuggingFace

- Developed a Persian-language Medicare chatbot powered by a Retrieval-Augmented Generation (RAG) framework. The dataset, meticulously curated from certified web sources, encompasses over 12,000 pharmaceutical questions and their corresponding answers. This chatbot delivers precise, context-aware responses, tailored to the unique needs of Persian-speaking users.

UTAI Hackathon | LinkedIn

- Developed an object detection model for vehicle damage assessment to support insurance claim automation during computer vision hackathon hosted by University of Tehran.

Dezhban | Restricted access (Available upon request)

- Built a hybrid model to identify suspected fraudulent drug prescriptions and claims using supervised and unsupervised learning algorithms. Using >20 specific algorithms it is capable of analyzing a large amount of data, it examines data distribution patterns, relationships between types of variables and their fluctuations over time, and identifies population subgroups based on commonalities analysis.

Torob Turbo | Restricted access (Available upon request)

- Built an LLM-based AI shopping assistant agent integrating retrieval-augmented generation for dynamic product recommendations.

SKILLS

Languages: Python | C++

Deep Learning Frameworks: PyTorch | Transformers | TensorFlow | Keras | timm | CUDA

Generative AI Frameworks: Pydantic-ai | Ollama | LangChain | LlamaIndex

Classical ML & Data science: Scikit-learn | XGBoost | Numpy | Pandas

Visualization: Plotly | Matplotlib | Seaborn | py3Dmol

Medical Imaging: MONAI | SimpleITK | Pydicom | BioPython

Dev & Deployment Tools: Git | Docker | MongoDB | ElasticSearch | Streamlit | Redis | Jupyter

AWARDS

Iran Biology Olympiad (IrBO) 2019

Silver medal

Iran Stem Cell And Regenerative Medicine Olympiad (ISRO) 2019

Top 0.1% nationally

National Olympiad of Entrepreneurship and Technological Startups 2024

Bronze medal

University of Tehran AI Hackathon (UTAI) 2025

3rd Place

+98 9192788837

a-mahdavikia@student.tums.ac.ir

Tehran, Tehran, Iran